

Part A [15 marks]

Question 1 [15 marks]

(a) [5 marks] Questions (i)-(v) are each worth one mark and have a single correct answer. If you believe more than one answer is correct, you should still only select one answer.

(i) Which of these numerical summaries is a measure of shape?

- A The sample standard deviation
- B The interquartile range
- C The sample skewness
- D The sample variance

(ii) Which of the following can always be determined from a relative frequency histogram of a sample of data?

- A The sample size
- B The sample mode
- C The sample standard deviation
- D None of the above

(iii) Which probability distribution would be most appropriate to model a continuous variate?

- A Multinomial
- B Binomial
- C Poisson
- D Exponential

(iv) The sample correlation may be an appropriate numerical summary for what kind of data?

- A Univariate data
- B Bivariate data
- C Both univariate and bivariate data
- D Neither univariate nor bivariate data

(v) If $R(\theta)$ is the relative likelihood function and $\hat{\theta}$ the maximum likelihood estimate, then what does $R(\hat{\theta})$ equal?

- A 0
- B 1
- C It depends on the value of $\hat{\theta}$
- D It depends on the value of θ

(b) [5 marks] Questions (i)-(v) are each worth one mark and have a single correct answer. You are not required to show any working for your calculations. No part marks are available.

A sample of size $n = 5$ contains the values $-4, -2, 0, 2, 4$. These data are stored in R in an object named `x`.

(i) What is the sample mean of `x`? 0

(ii) Is the sample standard deviation of `x` positive, negative, or zero? Positive

Note: This does not require calculation. Standard deviation cannot be negative, and will only equal 0 if all sample values are identical.

(iii) Is the sample skewness of `x` positive, negative, or zero? Zero

Note: This does not require calculation. The sample is perfectly symmetric, so the sample skewness must equal 0.

(iv) What number would be output if you ran the command `sum(x)`? 0

(v) Give *one* reason why it is impossible these data were sampled from a Poisson distribution.

There are negative values.

Note: An answer that suggests these data do not look like they fit the distribution of a Poisson model is not correct, as the question explicitly asks why it is *impossible* these data arose from a Poisson model.

(c) [5 marks] Questions (i)-(v) are each worth one mark and have a single correct answer. You are not required to show any working for your calculations. No part marks are available.

Suppose $y_1, y_2, \dots, y_{50}$ is an observed random sample from a probability distribution with corresponding independent and identically distributed random variables $Y_1, Y_2, \dots, Y_{50}$. You are told that the minimum value in the sample is 1, the maximum value in the sample is 10, and that:

$$\sum_{i=1}^{50} y_i = 100 \quad \sum_{i=1}^{50} (y_i - \bar{y})^2 = 600$$

where $\bar{y}$ denotes the sample mean.

(i) Suppose $y_1, y_2, \dots, y_{50}$ is an observed random sample from a $G(\mu, \sigma)$ distribution. What is the maximum likelihood estimate of μ ?

$$\frac{100}{50} = 2$$

(ii) Suppose that $Y_1, Y_2, \dots, Y_{50} \sim G(\mu, \sigma)$, independently. What is the maximum likelihood estimate of σ^2 ?

$$\frac{600}{50} = 12$$

Note: This question asks for the maximum likelihood estimate of σ^2 , not the sample variance. See notes in the lecture materials and (e.g.) a similar question on the practice midterm.

(iii) Suppose that $Y_1, Y_2, \dots, Y_{50} \sim \text{Poisson}(\theta)$, independently. What is the maximum likelihood estimate of θ ?

The sample mean, i.e., 2

(iv) Suppose that $Y_1, Y_2, \dots, Y_{50} \sim \text{Poisson}(\theta)$, independently. What is the maximum likelihood estimate of the standard deviation of Y_5 ?

$$\sqrt{2}$$

Note: The standard deviation of a $\text{Poisson}(\theta)$ distribution is $\sqrt{\theta}$, so via the invariance property we have that the m.l.e. is $\sqrt{\hat{\theta}}$.

(v) Suppose that $Y_1, Y_2, \dots, Y_{50} \sim \text{Poisson}(\theta)$, independently. What is the maximum likelihood estimate of $E \left[\sum_{i=1}^{20} Y_i \right]$? Hint: $\sum_{i=1}^n Y_i \sim \text{Poisson}(n\theta)$.

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Note: This is asking for the m.l.e. of the expected value of a sum of 10 i.i.d. $\text{Poisson}(\theta)$ random variables, which have expectation 10θ . So by the invariance property the answer is $10\hat{\theta}$.

Question 2 [15 marks]

(a) [5 marks] Questions (i)-(v) are each worth one mark and have a single correct answer. If you believe more than one answer is correct, you should still only select one answer.

(i) What type of variate is `vehicle.colour`?

- A Continuous
- B Discrete
- C Categorical
- D Ordinal

(ii) Which of these R commands would generate a scatterplot with `subject.age` as the response variate and `time.hour` as the explanatory variate for the assignment dataset?

- A `plot(mydata$subject.age, mydata$time.hour)`
- B `plot(mydata$time.hour, mydata$subject.age)`
- C Both A and B could be used.
- D Neither A nor B could be used.

(iii) Which of these R functions should be used when drawing a Gaussian probability density function curve on a relative frequency histogram?

- A `dnorm()`
- B `pnorm()`
- C `rnorm()`
- D `qnorm()`

(iv) How many traffic stops in the dataset in New Orleans happened on a Monday?

- A 12.5
- B 49
- C 349
- D 634

(v) What is the output of running the command

```
sum(mydata$subject.age.log)
```

- A [1] 3.495931
- B [1] 35.25895
- C [1] 3321.135
- D [1] 33496

Notes on Question 2 (a):

- (i) This follows directly from the definitions of variate types.
- (ii) The explanatory variate should be on the x -axis, and therefore is put first in the call to `plot()`, as covered on R Tutorial 1 and required for completing the first assignment.
- (iii) `dnorm()` plots the Gaussian probability density function, as covered in the Introductory R Tutorial, R Tutorial 1, and required for the first assignment.
- (iv) This can be deduced from knowledge of the assignment dataset. It cannot be 12.5 as that is not an integer. The total number of stops in New Orleans is 316, so the answer cannot be 349 or 634. The number of stops in New Orleans is included in your results from Assignment 1.
- (v) Even without knowledge of precise numbers, this value can be approximated from knowledge of the sample size (approximately 1000) and the sample mean of `subject.age.log` (approximately 3), both of which should be present in your answers to Assignment 1.

(b) [5 marks] Analysis 2c in Assignment 1 was the following:

- 2c. Generate an empirical cumulative distribution function plot with a suitable superimposed Gaussian cumulative distribution function curve for each of `subject.age` and `subject.age.log` for San Antonio. Your Gaussian cumulative distribution function curves should be a different colour to the e.c.d.f.

A student attempts to answer this question. You may assume they have correctly created an object named `SanAntonio` that contains only the traffic stops from San Antonio. The student then used the following R commands to answer Analysis 2c:

```
> mu <- median(SanAntonio$subject.age)
> sigma <- sd(SanAntonio$subject.age)
> plot(ecdf(SanAntonio$subject.age), xlab = "subject.age.log", main = "ecdf")
> curve(dnorm(x, mu, sigma))
```

Identify five mistakes the student has made in their R code. For each mistake, you should briefly describe what problem would arise in the resulting output. You do *not* need to explain how the code could or should be corrected, but you may do so if you feel it aids in your explanation. Note that simply providing a corrected R command is not sufficient to receive full credit: your answer must describe why the preceding code does not produce the desired output. Note: The plot's title is simply 'ecdf', this is *not* a mistake for the purposes of this question.

1 mark for each of the following, up to a maximum of 5.

1. This only generates a `subject.age` plot, the question requires a `subject.age.log` plot as well.
2. The `xlab` label is `subject.age.log`, but this is a plot of `subject.age`. Note: it is equivalently correct to say that it should be a plot of `subject.age.log` to match the axis label.
3. `dnorm()` is used, it should be `pnorm()`.
4. The `mu` term should be the mean, not the median.
5. The option `add = TRUE` should be in the `curve` call.
6. The curve should be a different colour.

Important Note: If you proposed errors that were not included in this list, you should still have received credit if they are indeed valid errors. If you gave an error that you believe was correct, but you did not receive credit, please follow the remark request procedures in the Midterm 1 Feedback LEARN announcement.

(c) [5 marks] Consider each of the following statements based on the assignment. For each statement, discuss whether you agree or disagree, providing a brief justification for your answer in 1-2 sentences.

(i) “Based on the sample correlation there is no relationship between `subject.age` and `vehicle.year` in San Antonio in the assignment dataset.”

Possible answer: Disagree. The sample correlation is 0.038. This is close to zero which indicates no linear association, but this does not necessarily mean there is no relationship of any kind.

Note: 2 mark question. The question does not make reference to a linear relationship, and so it is important to note in our answer that we cannot conclude there is no relationship (i.e., we must disagree with the statement); we can only make comments about a linear relationship. An answer of the form “We cannot conclude whether there is no relationship based solely on the sample correlation, as this only informs us of linear relationships” would also receive full credit.

(ii) “Because the sample mean of `subject.age` is different to the sample median of `subject.age` we can conclude that a Gaussian model is not appropriate for `subject.age` in San Antonio.”

Possible answer: Disagree. We cannot base such a conclusion on only one measure of comparison.

Note: 2 mark question. Many variations on this answer could be given. The key point for this statement is that when assessing the fit of a model we must consider a variety of model checking approaches, as we explored in Assignment 1.

(iii) “The median vehicle colour in San Antonio in the assignment dataset is white.”

Possible answer: Disagree. The concept of a median colour does not make sense.

Note: 1 mark question. This was part of Analysis 3 in Assignment 1.

Part C

Question 3 [10 marks]

Alex and Blake are two students working on co-op for *Goose River Transit*, a public transit company that provides bus services to residents of an area called Goose River. 12 months ago, the company introduced the *FlapPass*, a digital payment method that can be used instead of cash. Passengers may still pay in cash if they wish, but the cash fare is \$4 per journey, while each journey costs \$3 using the FlapPass.

The company conducted a survey of passengers while they were waiting for buses at the Goose River Transit Centre. A total of 100 passengers provided a response to the survey. Alex and Blake have been asked to analyze the resulting data, a summary of which is given in the table below.

Variate	Description	Sample Statistics	
Age	The passenger's age, in years.	Mean: 50.0	SD: 21.5
Journeys	The number of journeys the passenger took in the previous seven days.	Range: 0-28 Median: 12	Mean: 12.3 SD: 4.0
FlapPass	Whether the customer owns a FlapPass ('yes' or 'no').	Yes: 50	No: 50

(a) [2 marks] Based on the preceding information, answer each of the following:

(i) What type of variate is FlapPass?

Categorical. (Also acceptable: 'binary'.)

Note: This is not an ordinal variate. There is no implied 'ordering' for binary variates such as these. We can often think of ordinal variates as being a categorized version of an underlying continuous variate. For example, drink sizes of small, medium, large reflecting underlying liquid volumes. There is no analogous argument for whether or not someone owns a FlapPass: they either do or do not.

(ii) What is the interquartile range of Journeys in the sample?

NEI

Note: This cannot be determined from the available information.

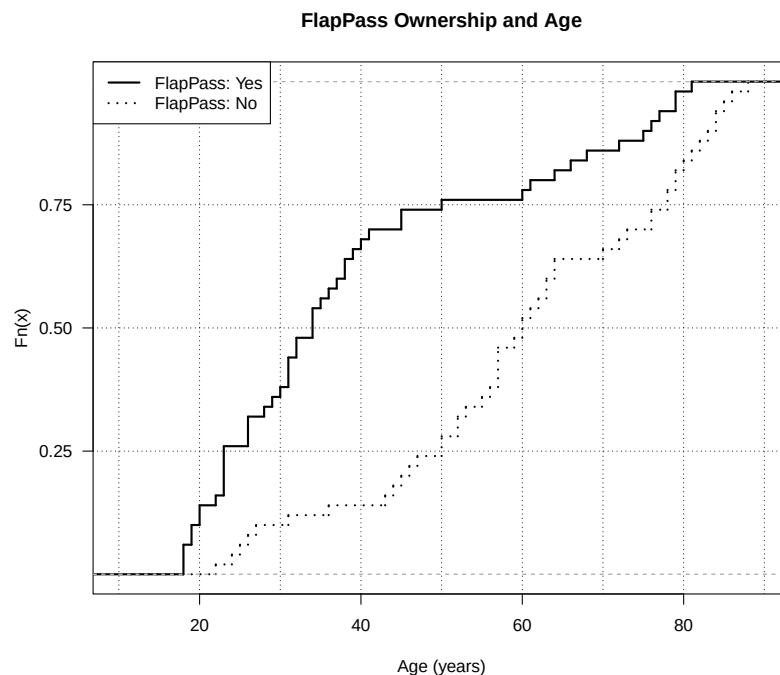
Let $j_1, j_2, \dots, j_{100}$ denote the number of journeys taken by each passenger in the sample. Assume these are realizations of a random sample $J_1, J_2, \dots, J_{100}$ where $J_i \sim \text{Poisson}(\theta)$, $i = 1, 2, \dots, 100$, independently.

(b) [3 marks] Give a numeric expression for the maximum likelihood estimate of the probability a randomly chosen passenger from the population took at least 14 journeys in the past seven days. Clearly indicate any results you use from the Course Notes.

We require $P(J \geq 14) = \sum_{j=14}^{\infty} \frac{\theta^j e^{-\theta}}{j!}$ and by the invariance property we have $\sum_{j=14}^{\infty} \frac{\hat{\theta}^j e^{-\hat{\theta}}}{j!}$ where $\hat{\theta} = 12.3$

Note: For full credit answers must include a correct numeric statement with all terms defined (including a specific numeric value for the m.l.e.) and reference the invariance property. Examples of this type were covered in class (Section 2.5) and Tutorial 2, as well as the end of chapter problems.

The plot below contains the e.c.d.f. for passenger age, split between those who have a FlapPass and those who do not. Horizontal and vertical dotted lines have been added to aid readability.



(c) [3 marks] *Estimate* a numeric value for each of the following. Your answer may be within ± 3 years of the correct value, except for part (iii) where your answer may be within ± 5 years of the correct value. Note: ‘not enough information’ is not a valid answer to any of these questions, even if you cannot calculate an exact answer, you should still estimate one.

(i) The age of the oldest passenger in the sample. 88

Note: Answers in the range 85-91 were accepted.

(ii) The median age of passengers in the sample who do not own a FlapPass. 60

Note: Answers in the range 57-63 were accepted.

(iii) The interquartile range of age of passengers in the sample who own a FlapPass. 25

Note: Answers in the range 20-30 were accepted). This can be estimated to the acceptable degree of accuracy by using the 0.25 and 0.75 quantiles on the y -axis.

(d) [2 marks] Blake tells Alex that because an equal number of people in the sample do and do not own a FlapPass, the mean age of those in the sample who own a FlapPass must be the same as those who do not own a FlapPass. Do you agree with Blake? Briefly explain why or why not.

No. We do not know that the distribution of number of journeys is the same between those who do and do not own a FlapPass.

Note: An argument could also be made based on the distribution evident in the e.c.d.f.

Question 4 [10 marks]

Charlie is the owner of a bubble tea and video game café called *MOBA Boba*. For several years the café has run a promotion on Tuesdays called *BOGO Boba* where customers can get two bubble teas for the price of one. Earlier this month, Charlie decided they wanted to study how long customers spend at the café, and whether they spend longer at the café during the *BOGO Boba* promotion.

Charlie took a sample of 30 customers from Monday September 8 and 30 customers from Tuesday September 16, and recorded the length of their visit rounded to the nearest second. Note that these were all the customers that visited the café on those days.

(a) [2 marks] Answer the following questions about the study.

(i) Is this an observational study or an experimental study? Observational [0.5 marks]

Note: This is not an experimental study because the BOGO Boba promotion had been in place for several years, but the decision to study customer behaviour was only made earlier this month. No changes were made by the experimenter to variates in the study.

(ii) Carefully define the units in the study.

Visitors to the café on Monday September 8 and Tuesday September 16. [1.5 marks]

Note: For full credit, the dates must be included in the definition of units. Also the question asks about the units in the study, not the units the researcher was interested in learning about.

Let $v_1, v_2, \dots, v_{60}$ denote the customer visit times in Charlie's sample. Assume these are realizations of a random sample $V_1, V_2, \dots, V_{60}$ where $V_i \sim G(\mu, \sigma)$, $i = 1, 2, \dots, 60$, independently.

(b) [2 marks] Based on the information given in the question above, give one reason why the proposed model is not appropriate. Your answer should include a brief justification.

Possible answer: The observations may not be independent, as customers may arrive in groups.

Possible answer: Visit times cannot be negative, whereas the Gaussian distribution extends past zero.

Possible answer: Because the samples are from two different days, it may not be appropriate to use a single probability model to describe them.

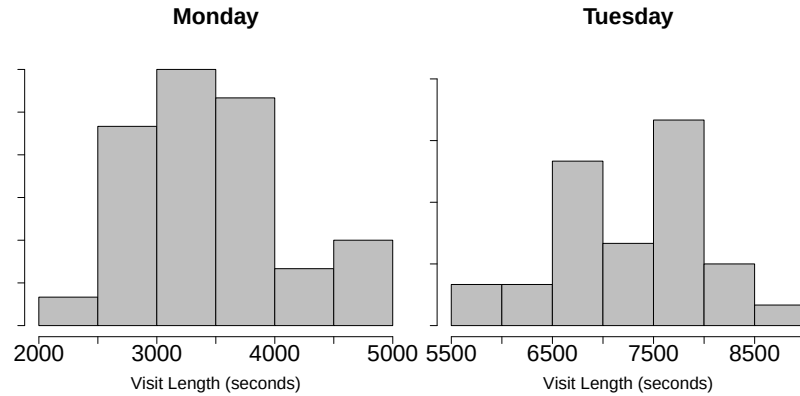
Note: The fact the times are rounded to the nearest second is *not* a reason why a Gaussian model is not appropriate. In all real-world settings we will have to round our data. We have seen examples in class (and in the first Assignment) where we have used continuous distributions to model data that are in some way rounded.

(c) [2 marks] Charlie explains to one of their employees that they have concerns about using a Gaussian model for these data. The employee suggests converting the times from seconds to hours by dividing by 3600 to try and improve the suitability of the Gaussian model. Do you agree or disagree with this advice? Justify your answer.

Possible Answer: Disagree. Changing the scale of a variate does not affect the shape of the distribution or any underlying concerns with the modelling assumptions.

Note: An argument that working in hours makes the Gaussian model less appropriate because hours might be viewed as 'more discrete' does not receive full credit, as this would only be a comprehensive argument if the times were rounded to the nearest hour. The question explicitly states the change of units was accomplished by dividing by 3600, and does not mention any rounding took place.

Charlie constructed relative frequency histograms of the visit times for the two days, as shown in the following figure. Note that the y -axis labels have been omitted.



(d) [2 marks] Based on the figure, for each of the following statements indicate whether it is True, False, or if Not Enough Information (NEI) is provided to determine if it is true or false.

Question	T/F/NEI
(i) The sample mean visit time on Tuesday was lower than the sample mean visit time on Monday. Note: All times on Tuesday were greater than 5500, all times on Monday were less than 5000.	FALSE
(ii) The sample median visit time across the entire sample of 60 customers was less than 6000 seconds. Note: The number of visits on each day is the same, therefore the median must be between the largest time on Monday and the smallest time on Tuesday. This question assesses understanding of the definition of median.	TRUE
(iii) The range of visit times on Tuesday was higher than the range of visit times on Monday. Note: The range for Monday is at least 2500-4500 and at most 2000-5000. The range for Tuesday is at least 6000-8500 and at most 5500-9000. We therefore cannot conclude whether the range for Tuesday is higher than the range for Monday. This question assesses understanding of how histograms are constructed, and what they can/can't tell us about a sample of data.	NEI
(iv) The modal visit time was between 3000 and 3500 seconds. Note: We cannot determine the mode from these histograms.	NEI

(e) [2 marks] Let s_m denote the sample standard deviation of visits on Monday, s_t denote the sample standard deviation of visits on Tuesday, and s_v denote the sample standard deviation of visits across the entire 60-visit sample. Which of the following statements is definitely true? Carefully justify your answer. If you believe it is not possible to determine if any of these statements are definitely true, then say so (with justification) in your answer.

$$s_v > s_m \quad s_v > s_t \quad s_v < s_m \quad s_v < s_t$$

We know $s_v > s_m$ and $s_v > s_t$. Because the two sets of data are disjoint (as indicated in the histograms) by combining the two samples the variance must be larger than the two samples separately.

Note: Only one of the two inequalities needed to be correctly identified. This question required careful justification, so to receive full credit it was important to clearly note that the two sets are disjoint.

Question 5 [10 marks]

A casino has a game which costs \$10 to play. Each play can result in one of three outcomes with an associated reward: a loss (\$0); a small prize (\$10); or a large prize (\$100). The casino states that the probability of winning a small prize is five times the probability of winning a large prize.

Let the random variable W represent the outcome of a random play of the game. We will state that W can take on the values 0, 10, or 100 (corresponding to losing, a small prize, or a large prize, respectively). We can write the probability function for W as follows:

$$P(W = w; \theta) = \begin{cases} 1 - 6\theta & w = 0 \\ 5\theta & w = 10 \\ \theta & w = 100 \end{cases}$$

and 0 otherwise. Note that the range of θ has been deliberately omitted.

You go to the casino and play the game 10 times. You win the large prize once, the small prize 4 times, and lose the remaining 5 times.

(a) [1 mark] Write down the possible range of θ .

$$0 < \theta < \frac{1}{6}$$

Note: Values of θ outside this range would result in probabilities that are less than 0 or greater than 1 in the probability function. The inequalities must be $<$ and not $\leq$: we know $\theta \neq 0$ and $\theta \neq \frac{1}{6}$ based on the observed data, as at least one large prize and one loss were observed. If $\theta = 0$, for example, then $P(W = 100) = 0$, which is inconsistent with the fact a large prize was won.

(b) [4 marks] Find the maximum likelihood estimate of θ based on these data. You may assume all plays of the game are independent.

Likelihood: [1 mark for correct general expression, 0.5 marks for substituting in probability function correctly]

$$L(\theta) = P(W = 0)^5 P(W = 10)^4 P(W = 100)^1 = (1 - 6\theta)^5 \cdot (5\theta)^4 \cdot \theta$$

Log-likelihood: [1 mark for taking logs correctly, award 0.5 marks if logs attempted and a minor mistake made]

$$\ell(\theta) = 5 \log(1 - 6\theta) + 4 \log(5\theta) + \log(\theta)$$

Differentiating: [1 mark for differentiating correctly, award 0.5 marks if differentiation attempted and a minor mistake made]

$$\begin{aligned} \ell'(\theta) &= \frac{-5 \times 6}{1 - 6\theta} + \frac{4 \times 5}{5\theta} + \frac{1}{\theta} \\ &= \frac{-30}{1 - 6\theta} + \frac{4}{\theta} + \frac{1}{\theta} = \frac{-30}{1 - 6\theta} + \frac{5}{\theta} \end{aligned}$$

Solving for 0: [0.5 marks for correctly solving for 0]

$$\begin{aligned} \frac{30}{1 - 6\hat{\theta}} &= \frac{5}{\hat{\theta}} \implies \frac{6}{1 - 6\hat{\theta}} = \frac{1}{\hat{\theta}} \\ \implies 6\hat{\theta} &= 1 - 6\hat{\theta} \implies 12\hat{\theta} = 1 \implies \hat{\theta} = \frac{1}{12} \end{aligned}$$

Note: Taking logs was not required, and full credit is available for a correct final answer if logs were not taken.

(c) [2 marks] What is the maximum likelihood estimate of the expected profit or loss for a single play of the game? You may write your answer in terms of $\hat{\theta}$. Hint: Begin by calculating $E[W]$. You are reminded the game costs \$10 to play.

$$E[W] = 0 \times (1 - 6\hat{\theta}) + 10 \times 5\hat{\theta} + 100 \times \hat{\theta} = 150\hat{\theta} = \frac{150}{12} = 12.5$$

So the player can expect to profit \$2.50, or $150\hat{\theta} - \$10$.

Part (d) concerns a different probability distribution, unrelated to parts (a)-(c).

Let $z_1, z_2, \dots, z_n$ be independent, identically distributed observations drawn from a continuous probability distribution with parameter θ . You are told that the probability density function for this distribution is:

$$g(z; \theta) = \theta z e^{z/\theta} \quad z > 0 \text{ and } \theta > 0$$

Note: This is not a real probability density function, but you may assume it has all the properties necessary in order to solve the following questions, and as such you may answer the following under the assumption it is a real probability density function. In particular, you may assume that the likelihood and log-likelihood functions are maximized for the value of θ at which their first derivatives equals 0.

(d) [3 marks] Find the maximum likelihood estimate of θ , expressed as a function of the observed data. Carefully show all your steps.

Likelihood:

$$\begin{aligned} L(z) &= \prod_{i=1}^n \left[\theta z_i e^{z_i/\theta} \right] = \theta^n \prod_{i=1}^n z_i \exp \sum_{i=1}^n z_i/\theta \\ &\propto \theta^n \exp \sum_{i=1}^n z_i/\theta \end{aligned}$$

$$\text{Log-likelihood: } \ell(z) = n \log(\theta) + \sum_{i=1}^n z_i/\theta$$

$$\text{Differentiating w.r.t. } \theta: \ell'(\theta) = \frac{n}{\theta} - \frac{\sum_{i=1}^n z_i}{\theta^2}$$

Solving for zero:

$$\frac{n}{\hat{\theta}} = \frac{\sum_{i=1}^n z_i}{\hat{\theta}^2}$$

$$\implies n\hat{\theta} = \sum_{i=1}^n z_i \implies \hat{\theta} = \frac{1}{n} \sum_{i=1}^n z_i$$